PROCUREMENT PACKAGE:	ME021 – Conveyor Weightometers
RIMERO PROJECT NO.	
EQUIPMENT NUMBER(S):	**EQUIPMENT NAME(S):**
F12-WD-402 F13-WD-403 F13-WD-405	STOCKPILE FEED CONVEYOR WEIGHTOMETER SAG MILL FEED WEIGHTOMETER RECYCLE CONVEYOR 2 WEIGHTOMETER
Description:	The Conveyor Weightometers will weigh incoming ore fed via conveyors to the Stockpile, SAG Mill, and Recycle Crusher.
Reference Specifications:	SPEC-00-G-05 – Technical Specification for Protective Coating 2000-F00-STS-GE-10001 – Packing and Shipping Specification 2000-F00-STS-GE-10002 – Site Specific Information 2000-F00-STS-EL-10001 – Electrical Equipment for Mechanical Packages Specification 2000-F00-STS-EL-10002 – Low Voltage Induction Motor Specification 2000-F00-STS-EL-10008 – Electrical Enclosures Specification 2000-F00-STS-EL-10014 – Preferred Equipment Suppliers for Electrical, Instrumentation and Controls Specification 2000-F00-STS-ME-10002 – Mechanical Equipment General Requirements Specification 2000-MEP-ME021-SCH1 – Vendor Document Schedule

SCOPE OF WORK

Work Included

Design, supply, and fabrication of three (3) Conveyor Weightometers including, but not limited to:

- Complete Conveyor Weightometer assembly including weighbridge, load cell(s), speed sensor and signal converter.
- Weigh quality idler sets, complete with lead in/lead out idlers.
- Field-mounted integrator.
- Digital speed sensor and associated electronics.
- IP65 enclosure.
- Electrical control panels with outputs to a remote control-room (panel design must be verified as AS 61439 compliant by the manufacturer who shall provide a statement of compliance).
- Calibration weights or chains as noted in the Data Sheet including storage frames.
- Automated deployment reeler for SAG Mill Feed Weightometer.
- All ancillary items and special tools normally supplied with or required to adjust and maintain all components associated with the equipment.
- All fasteners, gaskets, packers, shims, and any other items necessary for site assembly.
- Certified lifting lugs and suspension points, clearly stamped with the load limit and test certificates.
- Stainless steel nameplates engraved with manufacturer, model, serial number, and equipment number.
- Surface treatment of proprietary items to manufacturer's highest standard suitable for operation in an outdoor wet process plant environment subjected to hypersaline process and wash-down water:
    - Weighbridges and idler frames to be galvanised.
    - Masses to be painted in accordance with SPEC-00-G-05.
- Priced recommended spares list.

Specifications:

- All equipment to be supplied in accordance with the above referenced specifications.

Documentation / Quality:

- Vendor data and documentation in accordance with 2000-MEP-ME021-SCH1.
- Shop assembly and testing to the greatest extent possible with accompanying documentation including but not limited to:
    - Dimensional checks.
    - Testing of electric motors, protection, and control devices.
    - Calibration of instruments.
    - Operation of manual, pneumatic or electric mechanisms for valves.
    - Noise testing.
    - Testing of all other auxiliaries to confirm functionality.
- Performance guarantees and testing (state extents).

Shipping:

- Packaging suitable for transport as per 2000-F00-STS-GE-10001.
- Delivery as nominated in the commercial documentation.

Work Excluded

Excluded from this scope of works shall be the following:

- Foundations and/or structural supports.
- Hold-down bolts, nuts, and washers at concrete foundations/structural steel.
- Supply of all power, control, and instrumentation cabling outside battery limits.
- Offloading the equipment at site.
- Installation and commissioning at site.

Battery Limits

The works covered by this datasheet extend up to, and include the following battery limits:

- Underside of Weightometer mounting frame.
- Top of Weightometer idler rollers.
- Underside of roller chain deployment reeler mounting points.
- Electrical terminals of all Vendor supplied equipment.

Design Requirements

Weightometer:

- Belt weigher shall be capable of operation between 0% and 110% of conveyor design capacity.
- Lead in/lead out idler assembly frames shall have the same bolt hole fixing centres as the carry idler assembly frames. Vendor to obtain the bolt hole centres from Principal for verification purposes (once available).
- Lead-in, lead-out and weigh idlers shall be spaced a minimum of 1m and a maximum of 1.5m from each other.
- All weightometers must perform within the accuracies specified in the datasheet.
- All carry side idler assemblies shall be supplied with stone guards to protect wing rollers outer bearing assemblies.
- Weightometer to be suitable for operation in a dusty, dirty, wet, corrosive environment.
- The following materials are prohibited:
    - Aluminium, Asbestos, Copper, Poly-Chlorinated Bi-Phenyls (PCB's), Zinc and their alloys including Brass.

Guards:

- Guarding of the equipment in accordance with AS 4024.
- All guards shall be capable of being removed without disturbing other parts of the equipment and where possible be a maximum weight of 30kg. If this is not practical, supplier will seek approval from the Project, subsequent to approval lifting lugs or eye bolts shall be supplied, refer subsequent clauses on lifting requirements.
- Where guards interfere with access to lubrication points, stainless steel tubing and fixed remote grease nipples shall be provided so that lubrication can be undertaken without removing the guard and while the equipment is operating.
- Where any instruments that are inside a guarded area are required to be hand resettable – access from outside the guarding shall be provided to allow the instrument to be reset without risk to the operator.
- Guards shall have sufficient space for drive & component adjustment.

Lifting:

- Where practical all assemblies and components shall be provided with suitable lifting lugs or eye bolts to allow for easy off-loading and installation.
- Lifting lugs shall be stamped with a unique ID, the Working Load Limit (WLL) and provided with a Certificate of compliance stating that the lug conforms to relevant Australian Standards.

DUTY SPECIFICATION	UOM	EPC DATA	SUPPLIER DATA	NOTES
Site Data				
Site Location	-	WA		1
Site Data	-	Refer to 2000-F00-STS-GE-10002 - Site Specific Information		1
Application Data				
Duty	-	Continuous		1
Location	-	Outdoors		1
Operating Hours per Day	hr/day	24		1
Operating Days per Week	days/week	7		1
Operating Weeks per Year	weeks/year	52		1
Utilisation	%	65.0, 93.4		1,4
Design Life	years	30		1
Expected Ambient Operating Temperature Range	°C	1.7 to 37.2 Design for -5 to 50		1

DUTY SPECIFICATION	UOM	STOCKPILE FEED CONVEYOR WEIGHTOMETER F12-WD-402	SAG MILL FEED CONVEYOR WEIGHTOMETER F13-WD-403	RECYCLE CONVEYOR 2 WEIGHTOMETER F13-WD-405	NOTES
Process Requirements					
Material	-	Primary Crushed Gold Ore	Primary Crushed Gold Ore	Gold Bearing Pebbles	1
Material Specific Gravity	-	2.90	2.90	2.90	1
Ore Bulk Density (Nominal)	t/m³	1.60	1.60	1.60	1
Ore Bulk Density (Design Range)	t/m³	1.60 – 1.90	1.60 – 1.90	1.60 – 1.90	1
Moisture Content	% w/w	1.3	1.3	5.0	1
Particle Size (F_{80})	mm	122	122	49	1
Particle Size (F_{100})	mm	229	229	95	1
Conveyor Data					
Nominal Capacity	dry t/h	2,634	1,833	305	1
Design Capacity	dry t/h	3,135	2,182	610	1
Belt Speed	m/s	2.6	2.6	2.2	1
Belt Width	mm	1500	1500	750	1
Carry Idler Trough Angle	°	35	35	35	1
Carry Idler Type	-	Plain Inline	Plain Inline	Plain Inline	1
Carry Idler Diameter	mm	152	152	130	1
Belt Splice Type	-	Fabric PN1250/4 3-Step	Fabric PN1250/4 3-Step	Fabric PN630/3 2-Step	1
Carry Idler Spacing	m	1.0	1.0	1.0	1
Weightometer Details					
Weightometer Location	-	Tail-end, after Tramp Magnet	After Dust Collector	Tail-end, after Tramp Magnet	1
Belt Tension Range at Weightometer Location	kN	58 – 73 (Running) 64 – 95 (Starting)	70 – 74 (Running) 91 – 103 (Starting)	17 - 19 (Running) 20 - 27 (Starting)	1
Weightometer Type	-	Two idler/four load cell fully floating weigh frame	Three idler/four load cell fully floating weigh frame	Dual idler / 4 load cell weigh frame	3
Manufacturer / Model	-	CST / PFS4-2 Non-Intrusive	CST / PFS4-3 Non-Intrusive	CST / PFS4-2 Non-Intrusive	2
Weightometer Operational Range	t/h	20-120% design capacity (627-3762 tph)	20-120% design capacity (436-2618 tph)	20-120% design capacity (122-732 tph)	2
Weightometer Inclination	°	2.5	0	5	1
Maximum Weightometer Accuracy	%	± 0.5	± 0.25	± 0.5	1
Calculated Weightometer Accuracy	%	± 0.36	± 0.22	± 0.48	2
Number of Lead-In Idlers	#	2	2	2	1
Number of Lead-Out Idlers	#	2	2	2	1
Lead-In and Lead-Out Idler Spacing	m	1.2	1.5	1.0	2
Number of Weigh Idlers	#	2	3	2	2
Weigh Idler Spacing	m	1.2	1.5	1.0	2
Idler Frames Material	-	Mild Steel	Mild Steel	Mild Steel	3
Weigh Frames Material	-	Mild Steel	Mild Steel	Mild Steel	3
Idler Frames Finish	-	Galvanised	Galvanised	Galvanised	3
Weigh Frames Finish	-	Galvanised	Galvanised	Galvanised	3
Number of Idler Rollers / Type	# / -	3 / Trough	3 / Trough	3 / Trough	3
Idler Roller Diameter	mm	152	152	127	3
Idler Roller Bearing Type	-	6308	6308	6205	3
Idler Roller Shell Material	-	Mild Steel	Mild Steel	Mild Steel	3

DUTY SPECIFICATION	UOM	STOCKPILE FEED CONVEYOR WEIGHTOMETER	SAG MILL FEED CONVEYOR WEIGHTOMETER	RECYCLE CONVEYOR 2 WEIGHTOMETER	NOTES
		F12-WD-402	F13-WD-403	F13-WD-405	
Calibration Weights Included	-	On board calibration weights / hand operated	Permanently installed roller chain calibration weight on automated deployment reeler	On board calibration weights / hand operated	1
Calibration Weight - Percentage of Maximum Loading Available	%	0, 50 and 100	0 and 100 Or, 0 and 30 (Optional)	0, 50 and 100	2
Number of Load Cells	#	4	4	4	2
Length of Load Cell Cable	m	5.0	5.0	5.0	2
Load Cell Junction Boxes Required / Material of Construction	-	Yes / 316 Stainless Steel	Yes / 316 Stainless Steel	Yes / 316 Stainless Steel	1
Load Cell Mass	kg	500 (each)	1000 (each)	125 (each)	2
Weigh Length	m	2.4	4.5	2.0	2
Weigh Length Weight	kg	804	1050	154	2
Detector Power Supply	V	240	240	240	3
Control Voltage	V	24	24	24	3
PLC Output 1 Comms	-	4-20mA Isolated Analogue O/P	4-20mA Isolated Analogue O/P	4-20mA Isolated Analogue O/P	1
PLC Output 2 Comms	-	2x Vault Free Contact	2x Vault Free Contact	2x Vault Free Contact	1
PLC Output 3 Comms	-	Modbus TCP / IP	Modbus TCP / IP	Modbus TCP / IP	3
Remote Display	-	Yes	Yes	Yes	1
Remote Display Enclosure Design	-	Sloped Canopy	Sloped Canopy	Sloped Canopy	1
Remote Display Enclosure Ambient Protection	-	BW04042/S	BW04042/S	BW04042/S	3
Remote Display Enclosure Material	-	316 Stainless Steel	316 Stainless Steel	316 Stainless Steel	1
Remote Display Enclosure Surface Treatment	-	X15 Orange	X15 Orange	X15 Orange	1
Remote Display Enclosure IP Rating	IP	65	65	65	1
Remote Display Enclosure Gland Plate	-	6mm 316 Stainless Steel	6mm 316 Stainless Steel	6mm 316 Stainless Steel	1
Tacho Speed Sensor / Qty	-	Yes / 1	Yes / 1	Yes / 1	1
Tacho Speed Sensor Type / Location	-	Spiral Cage with 2 Mounts / Under carry belt	Spiral Cage with 2 Mounts / Under carry belt	Spiral Cage with 2 Mounts / Under carry belt	1
Tacho Speed Sensor Make / Model	-	IFM 2 wire / Rectangle	IFM 2 wire / Rectangle	IFM 2 wire / Rectangle	2
Local Control Panel / Junction Box					
Manufacturer / Make / Model	-	B&R BW40402/S, Jbox KKE030210FRS 1.5mm 316SS	B&R BW40402/S, Jbox KKE030210FRS 1.5mm 316SS	B&R BW40402/S, Jbox KKE030210FRS 1.5mm 316SS	2
Materials of Construction	-	316 Stainless Steel	316 Stainless Steel	316 Stainless Steel	1
IP Rating	IP	65	65	65	1
Surface Protection					
Preparation	-	Blast	Blast	Blast	2
Primer Type and DFT	µm	Dulux Durebild STE 150 µm	Dulux Durebild STE 150 µm	Dulux Durebild STE 150 µm	2
Top Coat Type, Colour and DFT	µm	Dulux Luxthane R 100 µm	Dulux Luxthane R 100 µm	Dulux Luxthane R 100 µm	2
Shipping Information					
Complete Assembly	kg	2121	4000 (including chain)	715	2
Idler Rollers (Each)	kg	18.5	18.5	10	2
Calibration Weight	kg	804	2099 plus support / enclosure	154	2
Pre-Assembly State - Describe Site Assembly Required	-	Minimised / packed for freight efficiency	Minimised / packed for freight efficiency	Minimised / packed for freight efficiency	2

Notes:

1. Data supplied by EPC The Supplier shall comply with this data.
2. Data shall be provided by Supplier.
3. Data proposed by EPC The Supplier shall confirm data or state alternative.
4. Stockpile Feed Conveyor to operate at 65.0% utilisation while SAG Mill Feed Conveyor and Recycle Conveyor to operate at 93.4% utilisation.